

Where irrigation exists is globally contested

2. Harmonization

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Contents

1	Harmonization	3
1.1	Does harmonization change the disagreement curves?	14
2	Session information	16

```

# REPROJECTING AND REGRIDDING #####
#####

sensobol::load_packages(c("data.table", "terra", "magrittr", "here", "rnaturalearth",
                          "sf", "countrycode"))

# LOAD ALL DATASETS #####

dt <- fread("./datasets/irrigated_areas/irrigated_areas.csv")

# Vector with name of datasets, used to build per-dataset rasters and stacks ---
maps <- unique(dt$dataset)

# Source all functions needed in the workflow -----
r_functions <- list.files(path = here("functions"),
                          pattern = "\\..R$", full.names = TRUE)
lapply(r_functions, source)

# Load countries as sf (medium scale is a good balance) -----
countries_sf <- ne_countries(scale = "medium", returnclass = "sf")

# Keep only what is needed -----
countries_sf <- countries_sf[, c("name", "geometry")]

```

1 Harmonization

```
# MAKE TEMPLATES #####

# Quick check that dataset totals look plausible and match expectations -----

dt[, sum(mha), dataset] %>%
  print()
```

```
##      dataset      V1
##      <char>    <num>
## 1:   gaez_v4 287.8335
## 2:     giam 343.1581
## 3:     gmia 253.0108
## 4:     luh2 267.7412
## 5:    meier 366.2890
## 6: mirca2000 216.0765
## 7:  mirca_os 271.3082
## 8:   nagaraj 297.2691
## 9:     spam 224.7258
## 10:   gripc 313.4176
```

```
# In terra, reprojection and aggregation are easier and safer when explicit
# template rasters defining extent, resolution and CRS are used. -----
```

```
template_ll_02 <- make_template_ll(0.2)
template_ll_04 <- make_template_ll(0.4)
template_ll_10 <- make_template_ll(1.0)
```

```
# Equal-area intermediate template (global Mollweide). -----
```

```
# Mollweide is a global equal-area projection suitable
# for conservation-style reprojection steps.
```

```
crs_ea <- "+proj=moll +lon_0=0 +datum=WGS84 +units=m +no_defs"
```

```
# Create an equal-area template with fixed world extents and 20 km resolution ---
```

```
# We selected this resolution because a fixed global extent avoids shifting bounds
# 20 km is fine enough to support global reprojection without excessive computation
```

```
template_ea <- rast(xmin = -18000000, xmax = 18000000, ymin = -9000000,
                    ymax = 9000000, crs = crs_ea, resolution = 20000)
```

```
# BUILD 0.2° STACK #####
```

```
# We selected 0.2° as a common support to preserve spatial heterogeneity and allow
# exact aggregation to coarser grids.
```

```

# Rasterize each dataset onto the 0.2° grid (area-preserving; totals conserved).
# make_raster_ll_02() implements an area-preserving rasterization step. -----

rast_ll_list_02 <- lapply(maps, make_raster_ll_02)
mha_stack_ll_02 <- rast(rast_ll_list_02)
names(mha_stack_ll_02) <- maps

# Check totals match original (post-rasterization) -----

# To ensure that the initial conversion from table to raster does not change totals.
orig_totals <- dt[, .(total_mha_orig = sum(mha, na.rm = TRUE)), dataset]

ll02_totals <- data.table(dataset = names(mha_stack_ll_02),
                          total_mha_ll02 = global(mha_stack_ll_02, "sum",
                                                    na.rm = TRUE)[, 1])

qa_ll02 <- merge(orig_totals, ll02_totals, by = "dataset") %>%
  .[, ratio_ll02 := total_mha_ll02 / total_mha_orig] %>%
  .[order(dataset)]

# Ratio_ll02 should be 1 for all datasets (or extremely close if floating error).
qa_ll02

## Key: <dataset>
##      dataset total_mha_orig total_mha_ll02 ratio_ll02
##      <char>      <num>      <num>      <num>
## 1:   gaez_v4    287.8335    287.8335      1
## 2:    giam     343.1581    343.1581      1
## 3:    gmia     253.0108    253.0108      1
## 4:   gripc     313.4176    313.4176      1
## 5:    luh2     267.7412    267.7412      1
## 6:   meier     366.2890    366.2890      1
## 7: mirca2000    216.0765    216.0765      1
## 8:  mirca_os    271.3082    271.3082      1
## 9:   nagaraj    297.2691    297.2691      1
## 10:   spam     224.7258    224.7258      1

# Export to .csv -----

fwrite(qa_ll02, "./datasets/irrigated_areas_regridded/QA_rasterization_totals_ll02.csv")

# FROM AREA TO FRACTION (0.2° LON / LAT) #####

# We convert to fraction because fractional coverage is bounded [0,1] and can be
# projected/resampled in a way that is conceptually consistent. It also avoids
# projecting "areas" directly in a CRS where cell areas vary with latitude.

```

```

A_ll02_ha <- cellSize(mha_stack_ll_02, unit = "ha") # cell area in ha
f_ll02 <- (mha_stack_ll_02 * 1e6) / A_ll02_ha # Mha to ha then divide by cell area to fraction

# Clamp to [0,1] to enforce physical constraints of fractional coverage -----

CLAMP_TO_01_OLD <- TRUE
if (CLAMP_TO_01_OLD) f_ll02 <- clamp01(f_ll02)

# REPROJECT VIA EQUAL-AREA: OPERATOR TEST (bilinear vs near) #####

# The bilinear operator -----

rt_old_bilin <- run_old_roundtrip(f_ll02 = f_ll02,
                                template_ea = template_ea,
                                template_ll_02 = template_ll_02,
                                method = "bilinear",
                                clamp_to_01 = CLAMP_TO_01_OLD)

# The near operator -----

rt_old_near <- run_old_roundtrip(f_ll02 = f_ll02,
                                template_ea = template_ea,
                                template_ll_02 = template_ll_02,
                                method = "near",
                                clamp_to_01 = CLAMP_TO_01_OLD)

mha_ll02_target_bilin <- rt_old_bilin$mha
mha_ll02_target_near <- rt_old_near$mha

# Quality assessment: total conservation (before renormalization) -----

tot_before <- global(mha_stack_ll_02, "sum", na.rm = TRUE)[, 1]
tot_bilin <- global(mha_ll02_target_bilin, "sum", na.rm = TRUE)[, 1]
tot_near <- global(mha_ll02_target_near, "sum", na.rm = TRUE)[, 1]

tot_check <- data.table(dataset = names(mha_stack_ll_02),
                        ratio_bilin = tot_bilin / tot_before,
                        ratio_near = tot_near / tot_before,
                        ratio_near_to_bilin = tot_near / tot_bilin) %>%
  .[order(dataset)]

print(tot_check)

##      dataset ratio_bilin ratio_near ratio_near_to_bilin
##      <char>      <num>      <num>      <num>
## 1:   gaez_v4   0.9897815  0.9876177      0.9978138

```

```
## 2:      giam    1.0005444  0.9999855          0.9994414
## 3:      gmia    0.9960749  0.9936816          0.9975973
## 4:      gripc   0.9951442  0.9932962          0.9981430
## 5:      luh2    0.9727439  0.9742801          1.0015792
## 6:      meier   1.0006563  1.0011468          1.0004901
## 7: mirca2000   1.0000218  0.9981450          0.9981232
## 8: mirca_os    0.9875943  0.9854969          0.9978762
## 9:   nagaraj   1.0024458  0.9992851          0.9968470
## 10:    spam    0.9953993  0.9938381          0.9984315
```

```
# Renormalize both operators to match original totals exactly -----
```

```
mha_ll02_target_bilin_cons <- renormalize_to_totals(mha_ll02_target_bilin, mha_stack_ll_02)
mha_ll02_target_near_cons  <- renormalize_to_totals(mha_ll02_target_near,  mha_stack_ll_02)
```

```
# Check exact conservation after renormalization (ratios should be 1) -----
```

```
tot_bilin_c <- global(mha_ll02_target_bilin_cons, "sum", na.rm = TRUE)[, 1]
tot_near_c  <- global(mha_ll02_target_near_cons,  "sum", na.rm = TRUE)[, 1]
```

```
qa_total_conservation_after_renorm <- data.table(dataset = names(mha_stack_ll_02),
          ratio_bilin_cons = tot_bilin_c / tot_before,
          ratio_near_cons  = tot_near_c  / tot_before) %>%
  .[order(dataset)] %>%
  print(.)
```

```
##      dataset ratio_bilin_cons ratio_near_cons
##      <char>          <num>          <num>
## 1:   gaez_v4             1             1
## 2:    giam              1             1
## 3:    gmia              1             1
## 4:   gripc              1             1
## 5:    luh2              1             1
## 6:    meier             1             1
## 7: mirca2000            1             1
## 8: mirca_os            1             1
## 9:   nagaraj            1             1
## 10:    spam            1             1
```

```
qa_total_conservation_after_renorm
```

```
##      dataset ratio_bilin_cons ratio_near_cons
##      <char>          <num>          <num>
## 1:   gaez_v4             1             1
## 2:    giam              1             1
## 3:    gmia              1             1
## 4:   gripc              1             1
## 5:    luh2              1             1
## 6:    meier             1             1
```

```

## 7: mirca2000      1      1
## 8:  mirca_os      1      1
## 9:   nagaraj      1      1
## 10:    spam      1      1

# SELECTION OF THE MAIN HARMONIZED PRODUCT #####
# We choose near because it preserves threshold-based presence/absence structure (no
# smoothing), and our quality assessment shows it minimally changes disagreement
# curves vs baseline.

mha_ll02_target <- mha_ll02_target_near_cons # MAIN
mha_ll02_target_sens_bilin <- mha_ll02_target_bilin_cons # SENSITIVITY

#Quality assessment threshold stability (before vs after) for BOTH operators ---

tau_set <- c(0, 100, 1000)

flip_old_bilin <- tau_flip_rates(mha_stack_ll_02,
                                mha_ll02_target_bilin_cons,
                                tau_ha = tau_set)

prev_old_bilin <- tau_prevalence_change(mha_stack_ll_02,
                                       mha_ll02_target_bilin_cons,
                                       tau_ha = tau_set)

flip_old_near <- tau_flip_rates(mha_stack_ll_02,
                                mha_ll02_target_near_cons,
                                tau_ha = tau_set)

prev_old_near <- tau_prevalence_change(mha_stack_ll_02,
                                       mha_ll02_target_near_cons,
                                       tau_ha = tau_set)

cat("\nFlip summary (OLD bilinear, conserved):\n")

##
## Flip summary (OLD bilinear, conserved):

print(flip_summary(flip_old_bilin))

##   tau_ha flip_up_median flip_up_min flip_up_max flip_down_median flip_down_min
##   <num>      <num>      <num>      <num>      <num>      <num>
## 1:    0    0.051292366 0.036123489 0.05842435    0.000000000 0.000000e+00
## 2:   100    0.019483329 0.015370487 0.02729641    0.001420646 3.125735e-06
## 3:  1000    0.007424245 0.006244592 0.01164461    0.001774167 8.983361e-04
##   flip_down_max
##   <num>
## 1: 1.812926e-05
## 2: 1.609753e-03

```

```
## 3: 2.069236e-03
```

```
cat("\nFlip summary (OLD near, conserved):\n")
```

```
##
```

```
## Flip summary (OLD near, conserved):
```

```
print(flip_summary(flip_old_near))
```

```
##      tau_ha flip_up_median flip_up_min flip_up_max flip_down_median
##      <num>          <num>          <num>          <num>          <num>
## 1:      0      0.001804799 0.0010489965 0.002209269      0.0018435582
## 2:     100      0.001616005 0.0010489965 0.001797923      0.0015831845
## 3:    1000      0.001024928 0.0008752057 0.001170275      0.0009830435
##      flip_down_min flip_down_max
##      <num>          <num>
## 1: 0.001065250      0.002159883
## 2: 0.001065250      0.001740409
## 3: 0.000821443      0.001147770
```

```
cat("\nPrevalence change (OLD bilinear, conserved):\n")
```

```
##
```

```
## Prevalence change (OLD bilinear, conserved):
```

```
print(prev_old_bilin)
```

```
##      dataset tau_ha      p_before      p_after      delta
##      <char> <num>          <num>          <num>          <num>
## 1:  gaez_v4      0 0.07292776 0.12265945 0.049731687
## 2:  gaez_v4     100 0.04775497 0.06560292 0.017847944
## 3:  gaez_v4    1000 0.02389311 0.02959133 0.005698214
## 4:    giam      0 0.04081209 0.09901389 0.058201802
## 5:    giam     100 0.03526641 0.05491791 0.019651493
## 6:    giam    1000 0.02280036 0.02941066 0.006610303
## 7:    gmia      0 0.07823026 0.12956670 0.051336439
## 8:    gmia     100 0.04481303 0.06158760 0.016774567
## 9:    gmia    1000 0.02268971 0.02808285 0.005393142
## 10: gripc      0 0.04998487 0.10839110 0.058406225
## 11: gripc     100 0.03657109 0.05046686 0.013895766
## 12: gripc    1000 0.02099493 0.02614052 0.005145584
## 13:  luh2      0 0.01479910 0.05617883 0.041379724
## 14:  luh2     100 0.01424210 0.04139660 0.027154506
## 15:  luh2    1000 0.01254232 0.02322296 0.010680635
## 16:  meier      0 0.05260299 0.10333803 0.050735048
## 17:  meier     100 0.04777560 0.06771029 0.019934685
## 18:  meier    1000 0.02904245 0.03522703 0.006184578
## 19: mirca2000      0 0.07232137 0.12421357 0.051892195
## 20: mirca2000     100 0.04337582 0.06144319 0.018067371
## 21: mirca2000    1000 0.02063735 0.02506151 0.004424165
## 22: mirca_os      0 0.07562465 0.12687294 0.051248293
```



```
## 23:  mirca_os      100 0.04589579 0.06169200 0.015796212
## 24:  mirca_os     1000 0.02287162 0.02780904 0.004937410
## 25:   nagaraj       0 0.03697369 0.07309718 0.036123489
## 26:   nagaraj     100 0.03697369 0.05851875 0.021545063
## 27:   nagaraj    1000 0.03083225 0.03747756 0.006645312
## 28:      spam       0 0.06681570 0.11836594 0.051550239
## 29:      spam     100 0.03942989 0.05597878 0.016548889
## 30:      spam    1000 0.01902885 0.02346676 0.004437918
##      dataset tau_ha  p_before    p_after      delta
##      <char>  <num>      <num>      <num>      <num>
```

```
cat("\nPrevalence change (OLD near, conserved):\n")
```

```
##
```

```
## Prevalence change (OLD near, conserved):
```

```
print(prev_old_near)
```

```
##      dataset tau_ha  p_before    p_after      delta
##      <char>  <num>      <num>      <num>      <num>
##  1:  gaez_v4       0 0.07292776 0.07287338 -5.438778e-05
##  2:  gaez_v4     100 0.04775497 0.04781249  5.751352e-05
##  3:  gaez_v4    1000 0.02389311 0.02400189  1.087756e-04
##  4:    giam       0 0.04081209 0.04087961  6.751587e-05
##  5:    giam     100 0.03526641 0.03532080  5.438778e-05
##  6:    giam    1000 0.02280036 0.02282599  2.563102e-05
##  7:    gmia       0 0.07823026 0.07820213 -2.813161e-05
##  8:    gmia     100 0.04481303 0.04479928 -1.375323e-05
##  9:    gmia    1000 0.02268971 0.02275097  6.126440e-05
## 10:   gripc       0 0.04998487 0.05003426  4.938661e-05
## 11:   gripc     100 0.03657109 0.03658547  1.437838e-05
## 12:   gripc    1000 0.02099493 0.02114872  1.537861e-04
## 13:   luh2       0 0.01479910 0.01480473  5.626322e-06
## 14:   luh2     100 0.01424210 0.01423084 -1.125264e-05
## 15:   luh2    1000 0.01254232 0.01256170  1.937955e-05
## 16:   meier       0 0.05260299 0.05258673 -1.625382e-05
## 17:   meier     100 0.04777560 0.04777373 -1.875441e-06
## 18:   meier    1000 0.02904245 0.02902745 -1.500353e-05
## 19: mirca2000     0 0.07232137 0.07232262  1.250294e-06
## 20: mirca2000    100 0.04337582 0.04336207 -1.375323e-05
## 21: mirca2000   1000 0.02063735 0.02064485  7.501763e-06
## 22:  mirca_os     0 0.07562465 0.07558464 -4.000940e-05
## 23:  mirca_os    100 0.04589579 0.04603457  1.387826e-04
## 24:  mirca_os   1000 0.02287162 0.02306730  1.956710e-04
## 25:   nagaraj     0 0.03697369 0.03695743 -1.625382e-05
## 26:   nagaraj    100 0.03697369 0.03695743 -1.625382e-05
## 27:   nagaraj   1000 0.03083225 0.03079286 -3.938426e-05
## 28:      spam     0 0.06681570 0.06677382 -4.188484e-05
## 29:      spam    100 0.03942989 0.03947053  4.063455e-05
```

```
## 30:      spam    1000 0.01902885 0.01908261 5.376263e-05
##      dataset tau_ha  p_before    p_after      delta
##      <char>  <num>      <num>      <num>      <num>

# SAVE QUALITY ASSESSMENT OUTPUTS FOR SUPPLEMENTARY MATERIALS #####

# These tables are evidence that harmonization does not bias results -----

fwrite(
  qa_total_conservation_after_renorm,
  "./datasets/irrigated_areas_regridded/QA_total_conservation_after_renormalization.csv"
)

fwrite(flip_old_bilin, "./datasets/irrigated_areas_regridded/QA_flip_rates_old_bilinear.csv")
fwrite(prev_old_bilin, "./datasets/irrigated_areas_regridded/QA_prevalence_change_old_bilinear.csv")
fwrite(flip_old_near,  "./datasets/irrigated_areas_regridded/QA_flip_rates_old_near.csv")
fwrite(prev_old_near,  "./datasets/irrigated_areas_regridded/QA_prevalence_change_old_near.csv")

# Sanity check to guarantee that the output is actually on the intended grid
# for downstream analysis -----

res(mha_ll02_target)

## [1] 0.2 0.2

# Verify totals after full round-trip vs ORIGINAL dt totals -----

ll02_target_totals <- data.table(dataset = names(mha_ll02_target),
                                total_mha_ll02_target = global(mha_ll02_target,
                                                                "sum", na.rm = TRUE)[, 1])

check_02 <- merge(orig_totals, ll02_target_totals, by = "dataset") %>%
  .[, ratio := total_mha_ll02_target / total_mha_orig] %>%
  .[order(dataset)]

print(check_02)

## Key: <dataset>
##      dataset total_mha_orig total_mha_ll02_target ratio
##      <char>      <num>          <num> <num>
## 1:  gaez_v4      287.8335        287.8335      1
## 2:   giam       343.1581        343.1581      1
## 3:   gmia       253.0108        253.0108      1
## 4:  gripc       313.4176        313.4176      1
## 5:   luh2       267.7412        267.7412      1
## 6:   meier       366.2890        366.2890      1
## 7: mirca2000     216.0765        216.0765      1
## 8:  mirca_os     271.3082        271.3082      1
## 9:  nagaraj      297.2691        297.2691      1
```

```

## 10:      spam      224.7258      224.7258      1
# SAVE THE ALIGNED 0.2° STACK #####

# We save the harmonized multi-layer raster for reproducibility and reuse.

writeRaster(mha_ll02_target, filename = "mha_stack_ll_02_aligned_NEAR.tif",
            overwrite = TRUE, gdal = c("COMPRESS=LZW"))

# SAVE THE OTHER RASTERS AS WELL #####

writeRaster(mha_stack_ll_02, "mha_stack_ll_02_baseline.tif", overwrite=TRUE, gdal="COMPRESS=LZW")

writeRaster(mha_ll02_target_bilin_cons, "mha_stack_ll_02_aligned_BILIN.tif", overwrite=TRUE, gdal="COMPRESS=LZW")

# Export 0.2° long table -----

irrigated_areas_regridded_02 <- as.data.table(
  as.data.frame(mha_ll02_target, xy = TRUE, na.rm = FALSE))

irrigated_areas_regridded_02 <- melt(irrigated_areas_regridded_02,
  id.vars = c("x", "y"),
  variable.name = "dataset",
  value.name = "mha",
  variable.factor = FALSE)

setnames(irrigated_areas_regridded_02, c("x", "y"), c("lon", "lat"))

# AGGREGATE TO 0.4° #####

# fact=2 maps 0.2° to 0.4°; summing preserves total irrigated area.

mha_stack_ll_04 <- aggregate(mha_ll02_target, fact = 2, fun = sum, na.rm = TRUE)
res(mha_stack_ll_04)

## [1] 0.4 0.4

# Check totals preserved under aggregation -----

orig_totals_stack <- global(mha_ll02_target, "sum", na.rm = TRUE)[, 1]
agg04_totals <- global(mha_stack_ll_04, "sum", na.rm = TRUE)[, 1]
qa_04 <- data.table(dataset = names(mha_ll02_target),
  ratio = agg04_totals / orig_totals_stack)

# Export assessment -----

fwrite(qa_04,
  "./datasets/irrigated_areas_regridded/QA_aggregation_totals_04deg.csv")

```

```

# Export 0.4° long table -----

irrigated_areas_regridded_04 <- as.data.table(
  as.data.frame(mha_stack_ll_04, xy = TRUE, na.rm = FALSE)) %>%
  melt(id.vars = c("x", "y"), variable.name = "dataset", value.name = "mha",
       variable.factor = FALSE)

setnames(irrigated_areas_regridded_04, c("x", "y"), c("lon", "lat"))

# AGGREGATE TO 1° #####

mha_stack_ll_10 <- aggregate(mha_ll02_target, fact = 5, fun = sum, na.rm = TRUE)
res(mha_stack_ll_10)

## [1] 1 1

# Check totals preserved under aggregation -----

agg10_totals <- global(mha_stack_ll_10, "sum", na.rm = TRUE)[, 1]
qa_10 <- data.table(dataset = names(mha_ll02_target),
                    ratio = agg10_totals / orig_totals_stack)

# Export quality assessment -----

fwrite(qa_10, "./datasets/irrigated_areas_regridded/QA_aggregation_totals_10deg.csv")

# Export 1° long table -----

irrigated_areas_regridded_10 <- as.data.table(
  as.data.frame(mha_stack_ll_10, xy = TRUE, na.rm = FALSE)) %>%
  melt(id.vars = c("x", "y"), variable.name = "dataset", value.name = "mha",
       variable.factor = FALSE)

setnames(irrigated_areas_regridded_10, c("x", "y"), c("lon", "lat"))

# ADD COUNTRIES + CONTINENT AND EXPORT #####

# 0.2° -----

irrigated_areas_regridded_02 <- add_admin(irrigated_areas_regridded_02)

## Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,
## Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,
irrigated_areas_regridded_02 <- drop_all_zero_or_na_cells(irrigated_areas_regridded_02)

# Final total check after dropping cells -----
# Even after dropping "all-zero" rows, dataset totals should remain unchanged.

```

```
merge(irrigated_areas_regridded_02[, .(current = sum(mha, na.rm = TRUE)), dataset],
      orig_totals, by = "dataset") %>%
  .[, fraction := current / total_mha_orig] %>%
  .[order(dataset)] %>%
  print()
```

```
## Key: <dataset>
##      dataset  current total_mha_orig  fraction
##      <char>    <num>         <num>    <num>
##  1:  gaez_v4 285.5135         287.8335 0.9919400
##  2:   giam 339.7342         343.1581 0.9900224
##  3:   gmia 249.8387         253.0108 0.9874624
##  4:  gripc 309.9096         313.4176 0.9888073
##  5:   luh2 265.5824         267.7412 0.9919371
##  6:   meier 362.3263         366.2890 0.9891815
##  7: mirca2000 213.6309         216.0765 0.9886819
##  8:  mirca_os 268.6011         271.3082 0.9900221
##  9:  nagaraj 293.5383         297.2691 0.9874498
## 10:   spam 222.3060         224.7258 0.9892322
```

```
# Export 0.2° file -----
fwrite(irrigated_areas_regridded_02,
       "./datasets/irrigated_areas_regridded/irrigated_areas_regridded_02.csv")
```

```
# 0.4° -----
irrigated_areas_regridded_04 <- add_admin(irrigated_areas_regridded_04)
```

```
## Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,
## Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,
irrigated_areas_regridded_04 <- drop_all_zero_or_na_cells(irrigated_areas_regridded_04)
```

```
fwrite(irrigated_areas_regridded_04,
       "./datasets/irrigated_areas_regridded/irrigated_areas_regridded_04.csv")
```

```
# 1° -----
irrigated_areas_regridded_10 <- add_admin(irrigated_areas_regridded_10)
```

```
## Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,
## Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,
irrigated_areas_regridded_10 <- drop_all_zero_or_na_cells(irrigated_areas_regridded_10)
```

```
fwrite(irrigated_areas_regridded_10,
       "./datasets/irrigated_areas_regridded/irrigated_areas_regridded_10.csv")
```

1.1 Does harmonization change the disagreement curves?

```
# DOES HARMONIZATION CHANGE THE DISAGREEMENT CURVES? #####

log_tau_min <- 0
log_tau_max <- 5
log_step <- 0.1

taus_ha_nonzero <- 10^(seq(log_tau_min, log_tau_max, by = log_step))
taus_ha <- c(0, taus_ha_nonzero)
taus_mha <- taus_ha * 1e-6

# Labels: "any_positive" for tau = 0, then e.g. "1ha", "1.26ha", ..., "100000ha"
tau_labels <- ifelse(taus_ha == 0, "any_positive",
                     sprintf("%gha", signif(taus_ha, 3)))

tau_grid <- data.table(tau_label = tau_labels,
                      tau_ha = taus_ha,
                      tau_mha = taus_mha)

# RELOAD PRODUCTS FOR THE QUALITY CHECK #####

mha_stack_ll_02 <- rast("mha_stack_ll_02_baseline.tif")
mha_ll02_target_near_cons <- rast("mha_stack_ll_02_aligned_NEAR.tif")
mha_ll02_target_bilin_cons <- rast("mha_stack_ll_02_aligned_BILIN.tif")

# Build disagreement curves for each product -----

curve_base <- compute_curve_from_stack(mha_stack_ll_02, tau_grid) %>%
  .[, product:= "base_ll02"]

curve_near <- compute_curve_from_stack(mha_ll02_target_near_cons, tau_grid) %>%
  .[, product:= "rt_near_cons"]

curve_bilin <- compute_curve_from_stack(mha_ll02_target_bilin_cons, tau_grid) %>%
  .[, product:= "rt_bilin_cons"]

curves <- rbindlist(list(curve_base, curve_near, curve_bilin), use.names = TRUE)
setorder(curves, product, tau_mha)

# Compute deltas vs baseline for key metrics -----

metrics <- c("frac_disagree", "frac_existence", "frac_marginal",
            "frac_minor_disagree", "frac_major_disagree",
            "share_existence", "share_marginal")

base_keep <- curve_base[, c("tau_label", "tau_ha", "tau_mha", metrics), with = FALSE]
```

```

delta_near <- merge(
  curve_near[, c("tau_label", "tau_ha", "tau_mha", metrics), with = FALSE],
  base_keep,
  by = c("tau_label", "tau_ha", "tau_mha"),
  suffixes = c("_near", "_base"))

delta_bilin <- merge(
  curve_bilin[, c("tau_label", "tau_ha", "tau_mha", metrics), with = FALSE],
  base_keep,
  by = c("tau_label", "tau_ha", "tau_mha"),
  suffixes = c("_bilin", "_base"))

# Convert to delta columns (harmonized - baseline) -----

for (m in metrics) {
  delta_near[, (m) := get(paste0(m, "_near")) - get(paste0(m, "_base")) ]
  delta_bilin[, (m) := get(paste0(m, "_bilin")) - get(paste0(m, "_base")) ]
}

delta_near[, comparison := "near_minus_base"]
delta_bilin[, comparison := "bilin_minus_base"]

delta_all <- rbindlist(list(delta_near, delta_bilin), use.names = TRUE, fill = TRUE)

# Summarize worst-case changes across tau for headline metrics -----

summary_near <- rbindlist(lapply(
  c("frac_disagree", "frac_existence", "frac_marginal"),
  function(m) summarize_delta(delta_near, m)
))[, comparison := "near_minus_base"]

summary_bilin <- rbindlist(lapply(
  c("frac_disagree", "frac_existence", "frac_marginal"),
  function(m) summarize_delta(delta_bilin, m)
))[, comparison := "bilin_minus_base"]

delta_summary <- rbindlist(list(summary_near, summary_bilin), use.names = TRUE)
setorder(delta_summary, comparison, metric)

print(delta_summary)

```

```

##           metric max_abs_delta tau_ha_at_max tau_label_at_max delta_at_max
##           <char>           <num>           <num>           <char>           <num>
## 1: frac_disagree 0.0332265432           0.0000    any_positive 0.0332265432
## 2: frac_existence 0.0332265432           0.0000    any_positive 0.0332265432
## 3: frac_marginal 0.0192117284           794.3282           794ha 0.0192117284
## 4: frac_disagree 0.0004796296          50118.7234          50100ha -0.0004796296

```

```
## 5: frac_existence 0.0001376543      630.9573      631ha 0.0001376543
## 6: frac_marginal 0.0004524691      50118.7234      50100ha -0.0004524691
##      comparison
##      <char>
## 1: bilin_minus_base
## 2: bilin_minus_base
## 3: bilin_minus_base
## 4: near_minus_base
## 5: near_minus_base
## 6: near_minus_base

# Save tables for supplementary materials -----

fwrite(curves,
       "./datasets/irrigated_areas_regridded/QA_disagreement_curves_base_near_bilin.csv")
fwrite(delta_all,
       "./datasets/irrigated_areas_regridded/QA_disagreement_curve_deltas_vs_base.csv")
fwrite(delta_summary,
       "./datasets/irrigated_areas_regridded/QA_disagreement_delta_summary.csv")
```

2 Session information

```
# SESSION INFORMATION #####

sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.3.1
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] countrycode_1.6.1   sf_1.1-0      rnaturalearth_1.1.0
## [4] here_1.0.2          magrittr_2.0.4  terra_1.8-93
## [7] data.table_1.18.2.1
##
```



```
## loaded via a namespace (and not attached):
## [1] cli_3.6.5          knitr_1.51          rlang_1.1.7         xfun_0.55
## [5] otel_0.2.0         DBI_1.3.0           KernSmooth_2.23-26  rprojroot_2.1.1
## [9] htmltools_0.5.9    e1071_1.7-17        rmarkdown_2.30      grid_4.5.2
## [13] evaluate_1.0.5     classInt_0.4-11     fastmap_1.2.0       yaml_2.3.12
## [17] compiler_4.5.2     codetools_0.2-20    Rcpp_1.1.1          rstudioapi_0.17.1
## [21] digest_0.6.39      class_7.3-23        tools_4.5.2         proxy_0.4-29
## [25] units_1.0-0

## Return the machine CPU -----

cat("Machine:      "); print(benchmarkme::get_cpu()$model_name)

## Machine:
## [1] "Apple M1 Max"

## Return number of true cores -----

cat("Num cores:    "); print(parallel::detectCores(logical = FALSE))

## Num cores:
## [1] 10

## Return number of threads -----

cat("Num threads: "); print(parallel::detectCores(logical = FALSE))

## Num threads:
## [1] 10
```